Shrinking the Constraint Envelope of the Voynich Manuscript: A Firewall-Disciplined, Self-Correcting Computational Program

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Abstract

We report a computational program on Beinecke MS 408 (the Voynich Manuscript) whose goal is not decipherment but the defensible *shrinking of the constraint envelope* around what the book is: which classes of encoding, language, and community of origin remain consistent with the evidence, with every claim carrying an explicit A–D evidence grade. Methodologically, the program couples three disciplines: a synthetic ground-truth *harness* (real-language, cipher, and null-generator controls) that every method must pass before touching the real manuscript; a *firewall* in which all statistics are produced by deterministic, versioned code writing machine-readable results; and a standing *clean-context adversarial refutation pass* applied to every finding. We confirm the well-known low-entropy anomaly (conditional character entropy $h_2 \approx 2.08$ vs 3.07–3.91 for medieval natural-language controls), a genuine two-system (Currier A/B) structure, strong positional grammar, and a dense paradigmatic morphology (edit-distance-1 main component 0.80). We find no lexical label→feature naming system for the herbal even at increased statistical power; verbose ciphers are excluded on morphology-network grounds while an abjad/abbreviation class is revived; and *no single statistic—nor a vector of them—localizes meaning*: against honest structured-meaningless baselines, a meaningless generator out-scores meaningful text on every axis we can measure. A candidate positive signal (a cross-organ root↔leaf visual regularity) was pursued to its measurement ceiling and is *unresolved and underpowered*. The most robust result is the architecture itself: across four iterations, the refutation pass repeatedly overturned the program's own conclusions—including one produced by its own analysis code—before they were reported. We make no translation or real-taxon claim of any kind.

Keywords: Voynich Manuscript; computational philology; constraint-based analysis; adversarial validation; evidence grading; reproducible research.

1 Introduction

The Voynich Manuscript (Beinecke MS 408) is an early-fifteenth-century codex written in an undeciphered script. Its computational study has a long record of confident, mutually incompatible "solutions," most of which pattern-match plausible output onto an unconstrained problem [Reddy and Knight, 2011, Bowern and Lindemann, 2021]. We take the opposite stance. We do not attempt to read the manuscript; we ask which *classes* of hypothesis the data can exclude or leave standing, and we require every claim to carry an evidence grade (A: replicated/robust; B: candidate; C:

directional/weak; D: speculative). Decipherment is explicitly not a goal, and no output of this program should be read as a translation.

The program's premise is that the field's recurring failure is methodological: plausible-sounding results are generated faster than they are refuted. We therefore invert the usual order—validation first, claims second—and we make the validation machinery, not any single finding, the primary contribution.

2 Data and Transliteration

We use the standard Zandbergen–Landini (EVA) transliteration as primary, with the v101 (GC) transliteration as a sensitivity layer [Zandbergen, 2024]. All text analyses stratify by the two Currier "languages" A/B [Currier, 1976] and by scribal hand [Fagin Davis, 2020]. Visual features of the herbal illustrations were annotated through a structured vision-model pipeline emitting coarse categorical morphology (e.g. root colouring, leaf arrangement) under a strict schema; annotation reliability is measured and carried through every downstream test. Natural-language comparanda are period-appropriate Latin, German, Italian, and (consonantal) Hebrew herbals and texts.

3 Method: Harness, Firewall, and Adversarial Refutation

Three disciplines are load-bearing.

Harness (validate before you claim). No method is trusted on the real manuscript until it behaves correctly on synthetic ground truth: real-language controls (H4), a verbose homophonic cipher (H2, Naibbe-class), a self-citation / auto-copying generator (H3) [Timm and Schinner, 2020], and parameterised null generators. A method that cannot separate these knowns is not used on the unknown.

Firewall (numbers come only from code). Every statistic is produced by deterministic, versioned code with fixed seeds, written to machine-readable JSON; each result records its producing script, git commit, dataset version, and parameters. No number in this paper is estimated or recalled—each is copied from a result file.

Adversarial refutation (a standing pass). Every finding is submitted to a clean-context skeptical reviewer instructed to build the strongest case against it, with positive/associational claims held to the harshest scrutiny (plausible-but-wrong positives being the field's main failure mode). A claim is graded A/B only after it survives this pass. As documented in §5, the pass has repeatedly overturned the program's own conclusions.

4 Results

4.1 Established structure [Grade A/B]

The conditional character-entropy anomaly replicates [A]. Voynichese second-order character entropy is $h_2 = 2.08$ (all words; 2.16 with spaces under the Lindemann–Bowern convention [Lindemann and Bowern, 2022]), far below matched medieval controls: German 3.07–3.22, Latin 3.15–3.23, consonantal Hebrew 3.91, while first-order entropy $h_1 \approx 3.87$ is unremarkable. The anomaly is robust across Currier A ($h_2 = 2.15$) and B (1.94) and across scribal hands.

Two statistically distinct systems (Currier A/B) [A]. Unsupervised clustering of pages by word co-occurrence recovers the Currier A/B division as the dominant structure (adjusted Rand index of the two largest clusters vs. Currier = 0.44 in EVA, 0.90 in v101)—the strongest structure recoverable from the text alone [Montemurro and Zanette, 2013].

A dense paradigmatic morphology [B]. The edit-distance-1 word network has a giant component covering 0.80 of types (Currier A 0.75, B 0.82), versus 0.16 for Latin and 0.22 for Italian—an order of morphological regularity unlike natural-language controls at matched size. Strong positional grammar (paragraph- and line-level regularities) likewise replicates.

Section↔text co-variation, Language A only [B]. Vocabulary tracks illustration section within Language A (within-dialect alignment ARI = 0.35) but not within Language B (0.004): the two systems differ not only in vocabulary but in how text relates to content.

4.2 What is disfavoured [Grade C]

No lexical naming system for the herbal [C]. An anchor hunt for word→feature associations returns no survivor, and a higher-power graded (count-based) re-run over the most powerful feature bands finds no association above chance (permutation $p = 0.48$ against a permuted-feature null; the raw multiple-testing "hits" are false positives). Label-adjacency anchoring is infeasible on the herbal (labels are page-unique). A power analysis bounds this: page-level anchors are recovered at effect size $\phi \geq 0.4$ but not below $\phi \leq 0.3$, so weak or rare-feature mappings are not excluded.

Verbose ciphers excluded; abjad/abbreviation revived [C]. A verbose cipher expands each plaintext letter into multiple glyphs, which *deductively* destroys the edit-distance-1 network (a Latin cipher's morphology main-component falls from 0.28 at 1:1 substitution to 0.001 under verbose expansion); homophone-rich verbose ciphers additionally collapse word-order information ($\Delta I = 0.013$ vs 0.356 for a type-preserving map). Conversely, an *abjad* of real Latin (vowel-dropping to consonantal skeletons) reaches the Voynichese morphology band (main component 0.90 vs the manuscript's 0.80)—so the dense network does not require a constructed morphology, and the abjad/abbreviation class [Hauer and Kondrak, 2016] is a live structural lead.

4.3 What is not established / underdetermined

Meaning is not localizable by structural statistics. We attempted to place the manuscript on interpretable meaningful↔meaningless axes. Against *honest structured-meaningless* baselines, the meaningless generator out-scores meaningful text on every axis: word-order information for a block drift-null is $\Delta I = 0.57$ vs 0.36 for natural Latin, and the morphology network for a meaningless abjad is 0.89 vs 0.87 for a constructed language—with the manuscript sitting *below both* endpoints ($\Delta I = 0.31$, morphology 0.80). Every VLM-computable statistic we tried is a structure detector, not a meaning detector. *The manuscript is maximally structured on every axis we can measure, yet no axis separates meaning.*

No encoding family is distinguished [C]. Under equal, held-out tuning with de-collinearised metrics, no generative family (cipher, self-citation, constructed language, abbreviation, abjad, and compositions thereof) is robustly closest to the manuscript; a whitened (Mahalanobis) re-analysis agrees but is itself regularisation-dependent (covariance condition number $\approx 1.0 \times 10^4$), which only reinforces that the encoding "bracket" is a descriptive ordering, not evidence for any family.

4.4 A candidate positive, resolved to unresolved: the root↔leaf bundle

The one associational signal that survived early scrutiny was a cross-organ visual regularity: a plant's root colouring co-varying with its leaf arrangement (Cramér's $V = 0.40$). Its history is a cautionary tale in miniature. A first re-annotation model failed to reproduce it ("single-model artifact"); a third model reproduced it and it survived stratification by scribal hand and dialect and a palette-style control (within-stratum permutation $p = 0.003/0.014$); we then added a genuinely out-of-lineage rater (a non-Anthropic frontier model whose root labels agree 0.86–0.91 with the others), and it did *not* reproduce the association—no cross-rater pairing survives. Crucially, the reproduce/not-reproduce split does not track vendor lineage, and the leaf feature is noisy for every rater (pairwise agreement ≈ 0.45). On the high-confidence consensus subset (raters agreeing on both organs) only $n = 74$ pages remain, with minimum detectable effect $\phi \approx 0.33$—underpowered for the ~ 0.28 effect, and where labels are reliable the association is $\approx$ nil ($V = 0.19$, $p = 0.63$). **The honest terminal status is unresolved: the feature is too noisy for model annotation to adjudicate; a pre-registered human panel is the only decisive test.** We record the whole arc, including its flip-flops, rather than report whichever model was added last.

5 Discussion

Two substantive conclusions and one methodological one. Substantively: (i) the manuscript is a genuine, richly structured, two-system formal script whose morphology network is reachable only by a length-preserving script over an already-structured inventory—which revives the abjad/abbreviation class and disfavours verbose ciphers; (ii) whether that structure carries *meaning* is underdetermined, and we show *why* it is hard: the statistics that quantify structure (word-order information, character entropy, morphological density) do not discriminate meaning, because structured-meaningless processes reproduce or exceed them. This reframes the central question from "what does it say" to "how far along each interpretable dimension does it sit," and marks the honest boundary at which purely distributional methods stop.

Methodologically, the architecture is the result. The clean-context refutation pass overturned, before publication, several of the program's own most attractive conclusions: a "meaning certificate" read of the word-order statistic; a "constructed-language best fit"; a "needs constructed morphology" claim (refuted by the abjad control the reviewer demanded); a "whitening confirms" claim (withdrawn once the covariance was shown ill-conditioned); a set of false-positive anchors caught only by a mandatory null control; and—twice—a positive root$\leftrightarrow$leaf verdict, first over-claimed and then over-killed, corrected to unresolved. A pipeline that declines to over-read a negative as readily as a positive, and that can overturn a verdict asserted by its own code, is, we argue, the transferable contribution to undeciphered-corpus research.

6 Limitations

Visual features are model-annotated and noisy (leaf-arrangement inter-rater agreement ≈ 0.45), which caps power on any imagery-based test; several conclusions rest on a single manuscript-length corpus and are correspondingly power-limited; all vision raters to date share broadly similar training lineages, so "independence" is cross-vendor, not absolute; and text results are primarily EVA-based, with v101 as a sensitivity layer rather than a full replication. Grades reflect these limits.

7 Future Work

The decisive next test for the root$\leftrightarrow$leaf question is a pre-registered independent *human* panel on the high-confidence consensus subset, with a power analysis reported first; more models cannot settle a noise-limited question. The main forward thrust is a *mid-level* linguistic program that builds the missing layers between structure and semantics—morpheme segmentation, distributional word-class (part-of-speech) induction, a function-word/content-word bimodality test, and syntactic dependency—run with Currier A/B as the primary stratification, to ask directly whether the two systems are two different generative processes. We also flag an abjad/abbreviation joint-signature test (can one parameterisation match entropy, word-order, and morphology simultaneously?) and a second cross-vendor annotator. None of these targets translation; each targets the grammar of the system, which is what the evidence can bear.

Reproducibility

All statistics are produced by deterministic, versioned code (fixed seeds) writing machine-readable JSON to `results/`; every reported number records its producing script, git commit, dataset version, and parameters. Source and data-processing code accompany this preprint.

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